

Closed-form 2nd-order Data Shapley as a Spurious-Correlation Attenuator

Problem We Tackle

ERM on **spuriously-correlated** data latches onto the majority *shortcut signal* (e.g. the color channel in Colored MNIST), collapsing minority / OOD worst-group accuracy.

Limitations of standard remedies.

- Bi-level meta-learning (FORML): per-minibatch inner SGD + implicit differentiation — implementation-heavy and costly.
- GroupDRO / IRM: requires **group labels** during training.
- JTT: 2-stage ERM, no intervention into the training *dynamics*.

Wang et al. 2024 derived a *closed-form* 1st-/2nd-order In-Run Shapley estimator from a single training run — but used it only for **post-hoc filtering**.

Our question: can the same closed-form formula be *repurposed as an in-training per-sample reweighter*, and does the 2nd-order cross-term contribute a **stand-alone benefit** over the 1st-order alone?

How We Approach It

Idea 1 — Post-hoc filtering → in-training weights.

Convert per-sample Shapley estimates into softmax weights every minibatch and run weighted SGD, instead of using them after training.

Idea 2 — Geometric meaning of the cross-term.

1st-order $\langle g_i, g_{\text{val}} \rangle$: gradient alignment.

2nd-order $\langle g_i, H_{\text{val}} g_{\text{val}} \rangle$: extra penalty on majority gradients aligned with *high-curvature directions of the validation loss* — curvature-aware down-weighting.

Idea 3 — RMS rescaling. The Hessian spectrum shifts by orders of magnitude during training; rescale the cross-term to the 1st-order’s RMS at every step to prevent divergence.

Idea 4 — EMA accumulation.

Per-step ϕ_i is noisy; smooth across iterations before applying softmax weighting.

- No implicit differentiation — single forward + backward + HVP.
- Group labels are used *only* to balance the anchor (never inside training).

Method: Fairds

Parameters θ_t , minibatch $\mathcal{B}_t$, anchor $\mathcal{D}_{\text{val}}$ (50:50 group-balanced). At each step:

- 1 **Per-sample** g_i via `vmap(grad)`.
- 2 **Validation** g_{val} + **Pearlmutter HVP** for $H_{\text{val}} g_{\text{val}}$.
- 3 **Closed-form Shapley score:**

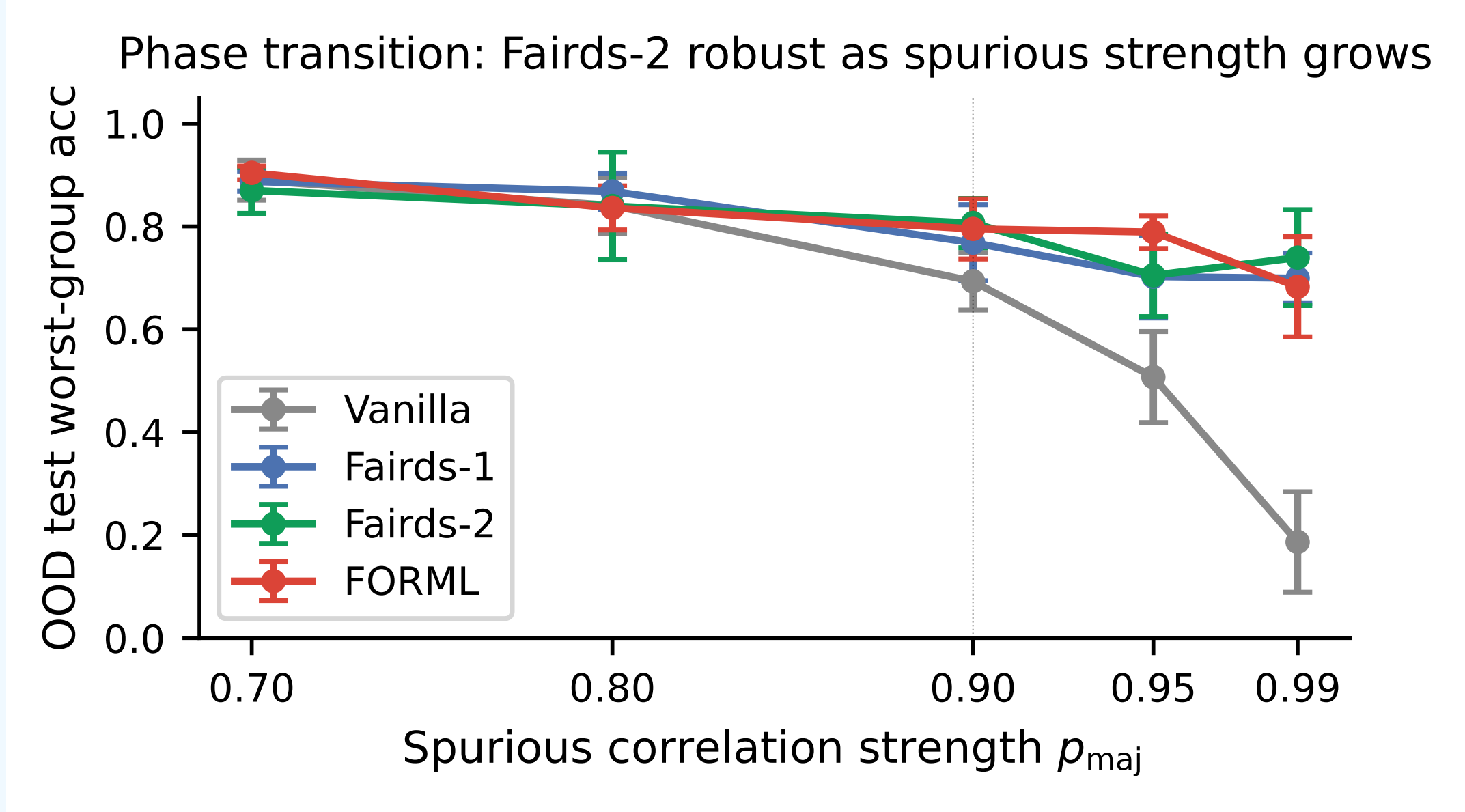
$$\begin{aligned}\phi_i^{(1)} &= \langle g_i, g_{\text{val}} \rangle, & \phi_i^{(2)} &= \phi_i^{(1)} - \alpha \tilde{c}_i \\ c_i &= \langle g_i, H_{\text{val}} g_{\text{val}} \rangle, & \tilde{c}_i &= c_i \cdot \frac{\text{RMS}(\phi^{(1)})}{\text{RMS}(c) + \epsilon}\end{aligned}$$

- 4 **EMA accumulate** $\bar{\phi}_i$ ($\beta=0.9$).
- 5 **Softmax-weighted SGD** with quantile clip.

Algorithmic contribution: RMS rescaling of the cross-term — without it the 2nd-order variant collapses under strong spurious bias.

Phase Transition: Spurious Strength Sweep

Sweep majority-correlation $p_{\text{maj}} \in \{0.7, \dots, 0.99\}$.



Vanilla collapses at $p=0.99$; Fairds-2 remains robust.

$p=0.95$: $\Delta = +19.8$ pp; $p=0.99$: $\Delta = +55.3$ pp ($p=3 \times 10^{-5}$).

Mechanism

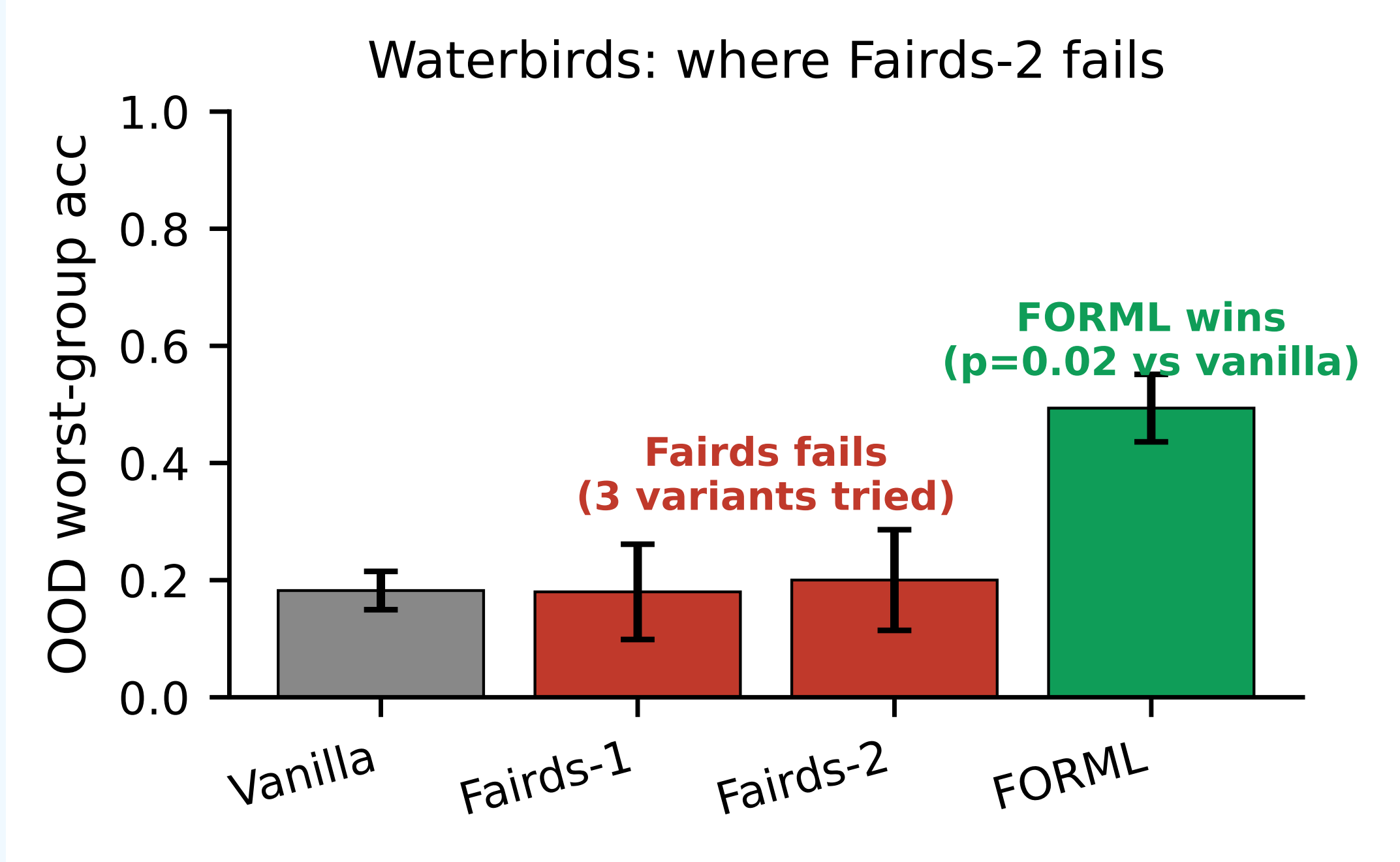
Controlled spurious-feature MLP, 2 of 8 input dims informative.

- Mann–Whitney $U(\text{maj} < \min \phi)$: $p=1.89 \times 10^{-8}$ at imbalance 0.9.
- $\Delta \phi$ monotone decreasing as imbalance grows.

Cross-term penalises majority gradients aligned with H_{val} ’s dominant directions — direct mechanism evidence.

Waterbirds: Regime Boundary

Pretrained ResNet-18 fine-tuning; 3 algorithmic variants tried.



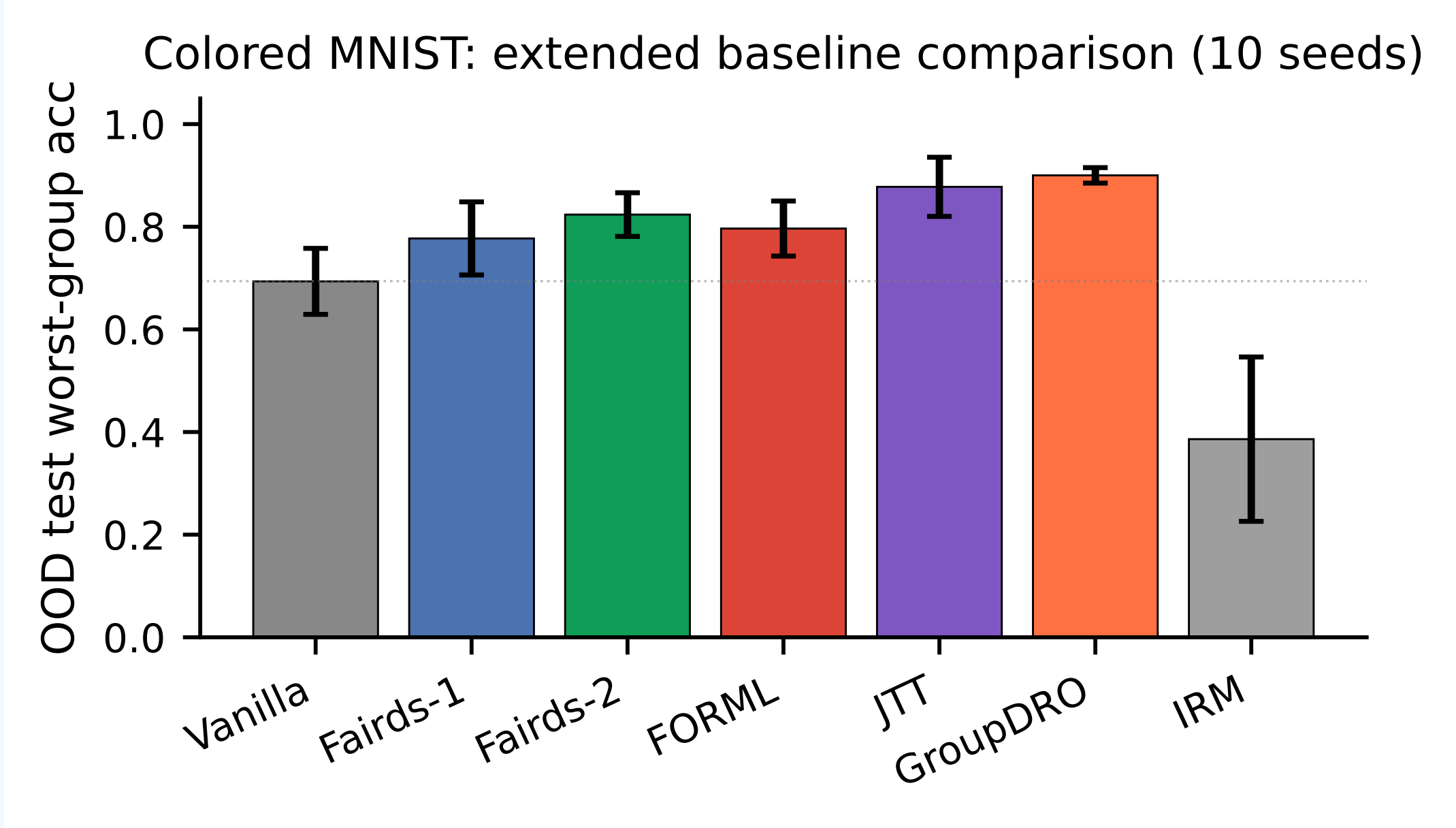
All Fairds variants $\leq$ Vanilla; FORML dominates.

Diagnosis: pretrained g_i is small / noisy in early epochs; EMA accumulates noise and reweighting destabilises training. No-EMA variant cannot recover — the underlying SNR is poor.

Recommendation: use **bi-level meta-learning** for pretrained fine-tuning regimes.

Colored MNIST: Held-out OOD

10 seeds, 7 methods, 20 epochs, best-tuned $\eta=0.05$.



Held-out OOD test_worst across methods.

Fairds-2 vs Vanilla: +13.0 pp ($p=5 \times 10^{-5}$)

Fairds-2 vs FORML: +2.7 pp ($p=0.052$)

Fairds-2 vs Fairds-1: +4.7 pp ($p=0.033$)

GroupDRO 0.900 and JTT 0.878 are stronger than Fairds-2 (0.824); we do not claim SOTA.

Takeaways

1. Closed-form 2nd-order Shapley reweighting is a **regime-dependent** tool.
2. Works on **from-scratch** / **spurious-correlation** regimes; the cross-term’s contribution is statistically isolated (+4.7 pp, $p=0.033$).
3. **Fails on pretrained fine-tuning** — bi-level meta-learning remains stronger.
4. **RMS rescaling** of the cross-term is the key to stability.